Yi-Chun Hung

Citizenship: Taiwan

Department of Physics, Northeastern University, Boston, MA, USA

Quantum Materials and Sensing Institute, Northeastern University, Burlington, MA, USA

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I. Research Summary

Theoretical condensed matter physicist specializing in flat-band physics, band topology, quantum geometry, and interaction-driven phenomena in low-dimensional and moiré systems. I combine analytical theory, numerical modeling, first-principles calculations, and symmetry-aware constrained generative models to investigate emergent quantum phenomena and advance quantum-materials discovery and design.

II. Education

Ph.D. in Physics • **Northeastern University** Sep. 2022 – present

M.S. in Physics • **Northeastern University** Aug. 2024

B.S. in Physics • **National Taiwan University** Jun. 2021

III. Research Experience

Doctoral Researcher • Ph.D. with Prof. Arun Bansil

May 2023 – present

Northeastern University, Boston, MA, USA

- Machine-learning-assisted materials design and discovery
- Lattice-geometry-driven flat-band physics
- Topological band theory beyond symmetry-protected topological phases
- Unconventional helical Luttinger liquids

Bachelor Researcher • Appointed by Dr. Hsin Lin

Sep. 2020 – Jun. 2022

Institute of Physics, Academia Sinica, Taipei, Taiwan

- Phenomenological superconducting proximity effect modeling

IV. List of Publications (*: co-first authorship)

1. **Yi-Chun Hung**, Ting-Wei Hsu, Qimin Yan, and Arun Bansil. Structurally Constrained Two-Dimensional Symmetric Crystal Generation for Quantum Materials Design. *(in preparation)*
2. **Yi-Chun Hung***, Yugo Onishi*, Hsin Lin, Liang Fu, and Arun Bansil. Extending Topological Bound on Quantum Weight Beyond Symmetry-Protected Topological Phases. ***Phys. Rev. Lett. 137***, 126601 (2026)
3. **Yi-Chun Hung**, Xiaoting Zhou, and Arun Bansil. Quantum Geometry Driven Finite-Momentum Exciton Fluctuations in Flat-Band Systems. *Materials Today Quantum,* 100076 (2026)

4. Baokai Wang*, **Yi-Chun Hung***, Xiaoting Zhou, Tzen Ong, and Hsin Lin. Feature Spectrum Topology. *arXiv:2310.14832* (accepted by Communications Physics)
5. **Yi-Chun Hung** and Arun Bansil. Helical Domain-Wall-Ring Networks Reshape Superconducting Correlations. *arXiv:2606.23599 (under review at PRB as a Letter)*
6. **Yi-Chun Hung** and Arun Bansil. Inter-Dirac-Cone Coherence Beyond the Valley Framework. *arXiv:2606.16155 (under review at PRL)*
7. **Yi-Chun Hung**, Xiaoting Zhou, and Arun Bansil. Twist-Induced Quantum Geometry Reconfiguration in Moiré Flat Bands. *arXiv:2603.20849 (under review at PRL)*
8. Xiaoting Zhou*, **Yi-Chun Hung***, and Arun Bansil. Quantum Geometry of Moiré Flat Bands Beyond the Valley Paradigm. *arXiv:2603.20852 (under review at PRL)*
9. **Yi-Chun Hung***, Tzen Ong*, and Hsin Lin. Nested Feature Spectrum Topology: Tripartite Topological Equivalence of Feature, Entanglement, and Wilson Loop Spectrum. *arXiv:2603.13128*
10. Mohammad Yahyavi, Ilya Belopolski, Yuanjun Jin, Yilin Zhao, Jinyang Ni, Naizhou Wang, **Yi-Chun Hung**, Zi-Jia Cheng, Tyler A. Cochran, Tay-Rong Chang, Wei-bo Gao, Su-Yang Xu, Jia-Xin Yin, Qiong Ma, Md Shafayat Hossain, Arun Bansil, Naoto Nagaosa, and Guoqing Chang. Wave-Function-Free Approach for Predicting Nonlinear Responses in Weyl Semimetals. *Phys. Rev. Lett. 136*, 046601 (2026)
11. **Yi-Chun Hung**, Chen-Hsuan Hsu, and Arun Bansil. Tunable Competing Electronic Orders in Double Quantum Spin Hall Superlattices. *Phys. Rev. B 112,* 195127 (2025)
12. **Yi-Chun Hung***, Xiaoting Zhou*, and Arun Bansil. Geometry-Driven Moiré Engineering in Twisted Bilayers of High-Pseudospin Fermions. *Phys. Rev. B 112,* L041403 (2025)
13. **Yi-Chun Hung**, Chen-Hsuan Hsu, and Arun Bansil. Majorana Kramers pairs in synthetic high spin Chern insulators. *Phys. Rev. B 111,* 245145 (2025)
14. Baokai Wang*, **Yi-Chun Hung***, Hsin Lin, Sheng Li, Rui-Hua He, and Arun Bansil. Topological Orbital Hall Effect. *Phys. Rev. B 111,* 195102 (2025)
15. Xiaoting Zhou*, **Yi-Chun Hung***, Baokai Wang*, and Arun Bansil. Generation of isolated flat bands with tunable numbers through Moiré engineering. ***Phys. Rev. Lett. 133***, 236401 (2024) ***Editors' Suggestion***
16. **Yi-Chun Hung**, Baokai Wang, Chen-Hsuan Hsu, Arun Bansil, and Hsin Lin. Time-Reversal Soliton Pair In Even Spin Chern Number Higher-Order Topological Insulators. *Phys. Rev. B 110,* 035125 (2024)
17. Yueh-Ting Yao, Xiaoting Zhou, **Yi-Chun Hung**, Hsin Lin, Arun Bansil and Tay-Rong Chang. Feature-energy duality of topological boundary states in a multilayer quantum spin Hall insulator. *Phys. Rev. B 109,* 155143 (2024)
18. Baokai Wang, Xiaoting Zhou, **Yi-Chun Hung**, Yen-Chuan Lin, Hsin Lin and Arun Bansi. High spin-Chern-number insulator in α-antimonene with a hidden topological phase. *2D Materials 11*, 025033 (2024)
19. Baokai Wang, **Yi-Chun Hung**, Xiaoting Zhou, Arun Bansil, and Hsin Lin. Higher-order topological phase hidden in quantum spin-Hall insulators. *Phys. Rev. B 108,* 245103 (2023)
20. Jing-Yang You, Xiaoting Zhou, Tao Hou, Mohammad Yahyavi, Yuanjun Jin, **Yi-Chun Hung**, Bahadur Singh, Chun Zhang, Jia-Xin Yin, Arun Bansil, and Guoqing Chang. Topological chiral kagome lattice. *arXiv:2309.00217*
21. Mohammad Yahyavi, Yuanjun Jin, Yilin Zhao, Zi-Jia Cheng, Tyler A. Cochran, **Yi-Chun Hung**, Tay-Rong Chang, Qiong Ma, Su-Yang Xu, Arun Bansil, M. Zahid Hasan, and Guoqing Chang. Transverse circular photogalvanic effect associated with Lorentz-violating Weyl fermions. *arXiv:2301.00958*

22. Peiling Li, Ya Deng, Chuang-Han Hsu, Chao Zhu, Jian Cui, Xue Yang, Jiadong Zhou, **Yi-Chun Hung**, Jie Fan, Zhongqing Ji, Fanming Qu, Jie Shen, Changli Yang, Xiunian Jing, Hsin Lin, Zheng Liu, Li Lu, and Guangtong Liu. Dimensionality-dependent type-II Weyl semimetal state in $Mo_{0.25}W_{0.75}Te_2$. *Phys. Rev. B 104,* 085423 (2021)

V. Analytical Expertise

- Quantum geometry and topological band theory
- Bosonization and perturbative RG analysis
- Band-projected interaction theory

VI. Numerical Skills

- Tight-binding calculations (lattice and Wannier-derived models)
- Continuum modeling for moiré materials
- Computing on HPC clusters (Explorer, NERSC)
- First-principles calculations using VASP
- Coding with MATLAB and Python 3
- Generative modeling for materials discovery

VII. Honors and Awards

1. Graduate Academic Excellence First-Year 2022-2023, Department of Physics, Northeastern University, Boston, MA, USA
2. Excellence in Teaching First-Year 2022-2023, Department of Physics, Northeastern University, Boston, MA, USA

VIII. Teaching Experience

Graduate Teaching Assistant

Sep. 2022 – May 2023

Northeastern University, Boston, MA, USA

Undergraduate Courses:

- Introductory Physics Laboratory (PHYS 1146 & PHYS 1148)
- Interactive Learning Section (PHYS 1153)

IX. Talks

1. "*Extending Topological Bound on Quantum Weight Beyond Symmetry-Protected Topological Phases*"
   - (poster) Gordon Research Conference on Correlated Electron Systems, South Hadley, MA, USA, Jun. 29 & 30, 2026
2. "*Quantum Geometry of Moiré Flat Bands Beyond the Valley Paradigm*"
   - (replacement speaker) APS Global Physics Summit, Denver, CO, USA, Mar. 19, 2026
3. "*Twist-Induced Quantum Geometry Reconfiguration in Moiré Flat Bands*"
   - APS Global Physics Summit, Denver, CO, USA, Mar. 16, 2026

4. "*More is different at the edge: helical Luttinger liquid in double quantum spin Hall insulators*"
   - (invited) Institute of Physics, Academia Sinica, Taipei, Taiwan, Feb. 10, 2026
5. "*Quantum Geometry Driven Fluctuation Phenomena in Flat-Band Systems*"
   - APS New England Section, Providence, RI, USA, Nov. 7, 2025
6. *"Generation of Isolated Flat Bands with Tunable Numbers through Moiré Engineering"*
   - (poster) APS New England Section, Providence, RI, USA, Nov. 7, 2025
7. *"Quantum Materials and Topology: Toward Robust Platforms for Quantum Computation"*
   - (invited) Annual Meeting of NSF-ExpandQISE 2: Exotic Physics and Applications of Solid-State Qubits, Arlington, VA, USA, Jul. 14, 2025
8. "*Emergent Majorana Kramers pairs in interacting spin/mirror Chern insulators*"
   - APS Global Physics Summit, Anaheim, CA, USA, Mar. 21, 2025
9. "*Moiré Engineering for Tunable Numbers of Flat Bands and Its Quantum Geometry*"
   - (invited) Institute of Physics, Academia Sinica, Taipei, Taiwan, Feb. 13, 2025
   - (invited) Department of Physics, National Cheng Kung University, Tainan, Taiwan, Feb. 19, 2025
10. "*Novel Moiré Pattern of a New Two-Dimensional Lattice*"
    - APS March Meeting, Minneapolis, MN, USA, Mar. 8, 2024
11. "*Time-Reversal Soliton Pair in Two-Dimensional Topological Insulating Systems*"
    - APS March Meeting, Las Vegas, NV, USA, Mar. 8, 2023
12. "*Edge Mass-Kink and Fractional Corner Charge in Higher-Order Topological Insulators*"
    - (invited) 1-day workshop on Wannier-functions based Hamiltonians 2022, National Center for Theoretical Sciences, Physics Division, Taipei, Taiwan, Apr. 29, 2022